DESIGN THINKING

Case Study

"Museum Collection Görlitz, Germany"

1 Empathize | 2 Define | 3 Ideate | 4 Prototype | 5 Test

My design thinking workflow follows a five-step process.

It begins with empathizing with users to understand their needs, followed by defining the core problem. From there, I move into ideation, where I explore multiple concepts and solutions. These ideas are then developed through prototyping and finally evaluated through testing.

Design Thinking — Phase

Empathize

1

1 Empathize

Context & Research Insights

While working on museum exhibitions, a recurring challenge became clear: families with children often disengage quickly from traditional, text-heavy formats.

This observation is supported by both professional experience (4 years working with museums) and research insights:

- A recent exhibition study (e.g. Picasso show) found that visitors spend on average only 4 seconds viewing an artwork, compared to 11 seconds in 2012.
- One possible explanation is the increasing influence of digital media consumption, leading to shorter attention spans.
- Studies also show that text length strongly affects engagement: Texts over 200 words are read by only 26% of visitors; texts around 100 words reach 59% readership.
- Additionally, larger visual objects attract longer attention, indicating the importance of visual hierarchy.

Key insight: Visitors scan rather than read, and engagement drops quickly without interaction or clear visual entry points.

Conclusion

Only 4 seconds of viewing per image is far too short

These insights suggest that one-size-fits-all exhibition formats are no longer effective. Instead, museum experiences need to:

- Adapt to shorter attention spans
- Provide layered content for different audience groups
- Prioritize visual engagement and interaction over long-form text

For families in particular, this means:

- Designing age-specific entry points (children vs. adults)
- Offering multiple ways to access content (visual, interactive, audio)
- Creating experiences that encourage active participation rather than passive reading

Project Context

This project started at a historical museum, where I was working on an exhibition aimed at a broad audience. One key challenge was that families with children often feel bored or disengaged during traditional museum visits. Standard exhibition formats and text-heavy content tend to be difficult for children to access, which can reduce overall engagement for the whole family.

Central question: How might we make museum exhibition topics more engaging and interesting for families with children?

Empathize Phase

As part of the Empathize phase, the focus was on understanding the visitors' experiences, frustrations, and needs. The goal was to feel the pain points of families with children during museum visits and to identify why traditional exhibition formats often fail to engage them.

These insights from the Empathize phase helped shape a clearer understanding of the problem and laid the foundation for the Define phase.

Design Thinking — Phase

Define

2

2
Define

Problem Statement

Families and children need an engaging and enjoyable museum experience that supports curiosity and interaction.

This will be successful when children actively participate in the exhibition and families spend more time engaging with the content.

Personas – Families Visiting the Museum

Child Persona 1: Max (8 years old, boy)

- Beliefs: Museums are boring if there's too much reading.
- Expectations: Wants to see, touch, and play; expects stories and activities to be fun.
- Frustrations: Loses interest quickly with text-heavy displays.
- Motivations: Curious about history when it is interactive; likes challenges and games.

Child Persona 2: Emma (6 years old, girl)

- Beliefs: Museums can be exciting if they have colorful visuals or interactive exhibits.
- Expectations: Wants visual storytelling, hands-on activities, and surprises.
- Frustrations: Feels bored when activities are slow or adult-oriented.
- Motivations: Enjoys discovering new things; likes to share discoveries with family.

Parent Persona 1: Sarah (Mother, 35 years old)

- Beliefs: Museums should be educational but also enjoyable for her children.
- Expectations: Wants her children to learn while having fun; expects clear signage and easy-to-follow paths.
- Frustrations: Feels stressed when children get bored or restless.
- Motivations: Wants a positive family experience; values both education and engagement.

Parent Persona 2: Tom (Father, 38 years old)

- Beliefs: Museums are best experienced when children are interested and engaged.
- Expectations: Wants interactive exhibits that allow family participation.
- Frustrations: Feels exhibits are too passive or adult-focused; wants quality time with family.
- Motivations: Seeks memorable experiences; values hands-on learning and fun for children.

"How Might We" Questions

For Children (Max & Emma)

- HMW make the exhibition fun and interactive, so children stay engaged?
- HMW use storytelling and visual elements to spark curiosity in children?
- HMW design hands-on activities that let children explore and learn actively?

For Parents (Sarah & Tom)

- HMW create a family-friendly experience that allows parents and children to enjoy together?
- HMW reduce frustration for parents when children lose interest?
- HMW make educational content enjoyable and accessible for both children and adults?

Combined / Overarching HMW

- HMW transform museum visits into a memorable experience for the whole family?
- HMW encourage interaction and participation across all age groups in the exhibition?
- HMW balance educational content with playful and engaging experiences?

Design Thinking — Phase

Ideate

3

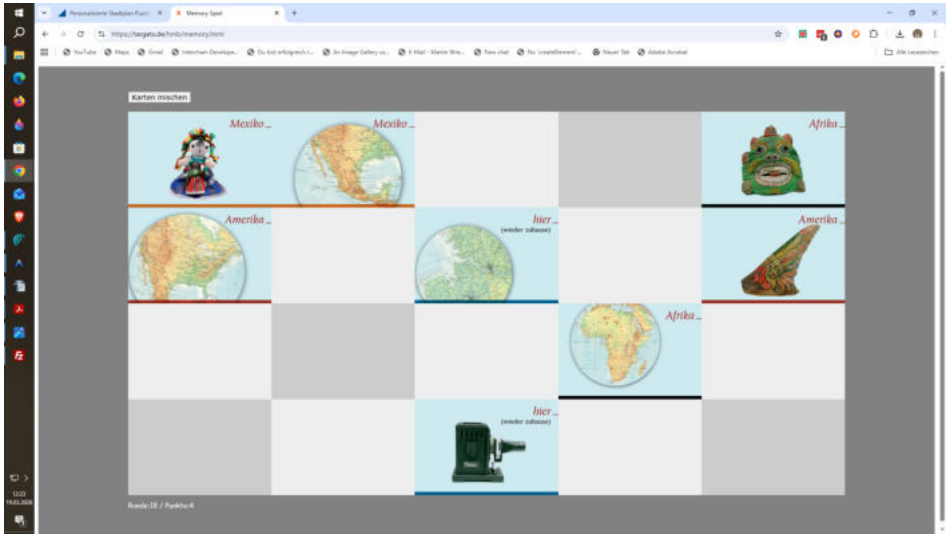
3
Ideate

Integration Concepts – Engaging Families

Augmented Reality and interactive experiences at museum exhibitions

- 1. Interactive Storytelling Zones – Children follow a narrative adventure; touchscreens, props, or AR let them discover objects and stories.
- 2. Hands-On Activity Stations – History puzzles, artifact reconstruction, or interactive map games; short, fun, shareable tasks.
- 3. Gamified Exhibition Path – A 'museum quest' with stamps, clues, and small rewards (stickers, certificates) to increase dwell time.
- 4. Visual & Multimedia Enhancement – Colorful graphics, animations, projection mapping; interactive panels for children to control.
- 5. Family Reflection / Sharing Spaces – Collaborative drawing boards, photo moments, or digital storytelling stations at the end of exhibits.
- 6. Individual Text Descriptions & Audio Guides – Different text versions for different visitor categories: adult, young adult, child (boy), child (girl).

Concept: Gamified Exhibition Path



Online Memory Game with the main topics of an exhibition

- A "museum quest" where children collect stamps, solve clues, or complete challenges.
- Parents can track progress and encourage participation.
- Integration with small rewards (stickers, certificates) for motivation.
- Benefit: Creates a playful structure that increases dwell time and attention.

Design Thinking — Phase

Prototype

4

4
Prototype

Prototype – Narrative Entry Point for Family Engagement

To explore how storytelling and interaction could increase engagement, a prototype was developed in the form of a narrative-driven exhibition poster.

The concept introduces a set of characters, each representing a different museum location (e.g. towers, museum houses, library). These characters act as guides through the exhibition, creating a cohesive story across multiple spaces.

Each character is linked to additional content via QR codes:

- Background stories
- Interactive elements (e.g. games or challenges)
- Extended information tailored to different age groups

Hypothesis & Problem–Solution Fit

Hypothesis

- Story-driven navigation increases engagement, especially for children.
- Visitors are more likely to explore multiple locations when guided by characters and narrative continuity.
- Interactive, on-demand content (QR codes) encourages deeper exploration compared to static text.

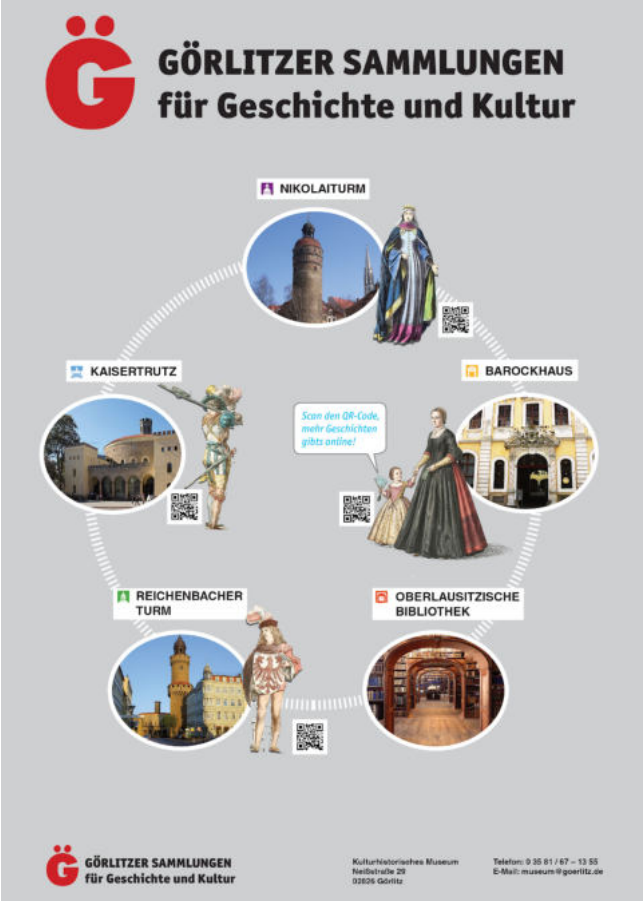
Problem–Solution Fit

- Short attention spans → replaced by bite-sized, story-based content
- Text-heavy exhibitions → complemented with interactive and visual storytelling
- Lack of engagement for children → solved through playful characters and exploration

Why this prototype

- Tests narrative engagement, cross-location experience, and digital interaction (QR codes) simultaneously.
- Remains low-cost and easy to implement, making it realistic for museum environments.

Exhibition Poster – Characters & Locations



Each historic character represents a museum location:
1 Nikolaiturm: Aunt 'Alex' · 2 Barockhaus: girl 'Emma' · 3 Reichenbacher Turm: teenage boy 'Max' · 4 Oberlausitzische Bibliothek: mother 'Sarah' · 5 Kaisertrutz: father 'Tom'

The QR code on the poster links to background stories of the places and each character — enabling stories about the history of places, games, and books sold in the bookstore.

Prototype – Narrative & Spatial Experience

A prototype was developed as a combined visual and spatial concept with two integrated elements:

- A narrative-driven poster system introducing characters that represent different museum locations
- A set of spatial visualizations (3D views) showing how these characters and interaction points could be placed within the museum and across the city

Together, they guide visitors through exhibitions using story, space, and interaction.

Concept

- Visitors encounter characters as physical elements (e.g. standees, signage)
- QR codes connect to additional digital content — stories, games, or deeper information
- The narrative continues across locations, encouraging exploration beyond a single exhibition space

3D Visualizations illustrate:

- Placement of characters at key decision points (entrances, transitions, highlights)
- Visibility and scale within exhibition spaces
- Interaction moments between visitors and the installation

Prototype in Context

Poster in front of the museum Kaisertrutz · Emma in the Barockhaus as a life-size cardboard cutout inside the historic rooms

Design Thinking — Phase

Test

5

5
Test

Initial Validation & Stakeholder Feedback

The concept was presented to museum stakeholders and city officials from the tourism department as part of an early validation phase.

Overall, the feedback was very positive. Stakeholders particularly appreciated the broad and strategic scope of the concept, which went beyond their initial expectations and introduced new ways of engaging families across multiple museum locations.

At the same time, the discussion revealed that some of the underlying assumptions about the specific context of the museum and the city of Görlitz were not fully accurate. This highlighted gaps in contextual and stakeholder-specific research.

Key Learnings

- Early stakeholder involvement is valuable to validate the strategic direction, even before user testing.
- Assumptions based on prior experience should be complemented with context-specific research.
- Aligning concepts with operational and local realities is critical for successful implementation.

Next Steps

- Deeper collaboration with local stakeholders
- More targeted research into the museum's visitors and city tourism dynamics
- Validation of concepts with actual families through usability testing